



GMUNC

Introduction

GMUNC will be held in person on October 19, 2024, at Henry M. Gunn High School (780 Arastradero Road, Palo Alto, CA, 94306). For more information about the conference, please visit the GMUNC website at gmunc.weebly.com. Delegates can also find guidelines for position papers on the website. The deadline for the position papers is October 12. Any submission after the due date is ineligible for any awards given out by the conference. October 18 is the deadline for position papers in order to be considered for research awards. If delegates are in need of an extension for position paper deadlines, would like to submit their position papers, or have questions or concerns, please contact: icaogmuncxi@gmail.com

Letter from Chair:

Dear Delegates,

Welcome to the International Civil Aviation Organization 2024! In this committee, we will discuss how to improve the reliability of avionics and operational safety to create a safer industry.

GMUNC welcomes delegates of all skill levels and will provide a review of standard parliamentary procedures at the beginning of the committee. We urge you to actively contribute to the committee and enthusiastically collaborate with other delegates. However, do be mindful that this conference may be the first MUN conference for some delegates, so please act with courtesy.

We encourage you to examine your delegation's stance through plenty of research and think about potential plans you would like to discuss throughout the conference. We also expect delegates to come prepared to compromise and face challenging discussions in these unprecedented and urgent times.

But most importantly, we hope you have a fun and engaging experience.

If you have questions regarding how to write your position paper, conference information, or any general questions, please don't hesitate to reach out to us at icaogmuncxi@gmail.com

Sincerely,

Eason Lu

Dolly Wu

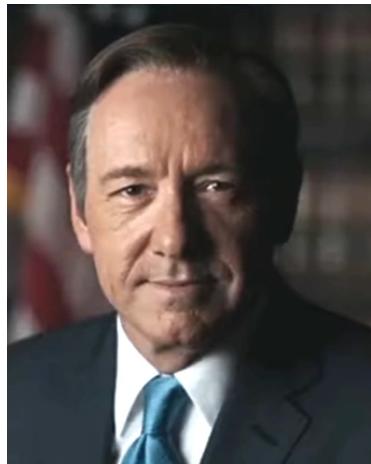
Meet your Chairs:

Eason Lu:



Eason Lu is a junior at Gunn High School and is excited to be serving as your chair this year. He has been participating in MUN for five years and has previously chaired GMUNC X. Outside of MUN, he is a certified pilot and radio operator. He is also a professional in emergency management, specializing in continuity of operations.

Dolly Wu:



Dolly Wu is a junior at Gunn High School and is serving as your co-chair this year. She has been participating in MUN for four years and has previously been crisis staff for GMUNC X. Outside of MUN, she is a varsity debater and is part of Gunn's Ethics Bowl team. She is also very active in helping improve youth civic engagement and advocating for youth mental health resources/policies. Lastly, she is a big music enthusiast!

Topic Introduction

In aviation, the safety and reliability of avionic systems and operational procedures are paramount. Avionics refers to the electronic systems that are used for communication, navigation, maintenance, monitoring, and flight control, and form the backbone of modern aviation, from takeoff to landing. The reliability of these systems directly influences the safety, efficiency, and most importantly the public presentation of aviation. Ensuring that these systems function properly without error is critical in preventing accidents, minimizing risks, reducing downtime, and most importantly: **avoiding the endless amount of paperwork required after even minor incidents.**

The International Civil Aviation Organization (ICAO) is a specialized agency of the United Nations that plays a crucial role in fostering the safe and orderly development of international civil aviation. Established in 1947, ICAO sets global standards and regulations necessary for aviation safety, security, efficiency, and environmental protection, by developing international standards and recommended practices (SARPs).

This background guide provides an overview of current situations regarding avionics reliability, and operational safety, including an examination of relevant terms, technology, and definitions. Historical accidents have shaped the current regulatory landscape, and this guide will include past accidents and steps taken to mitigate such accidents.

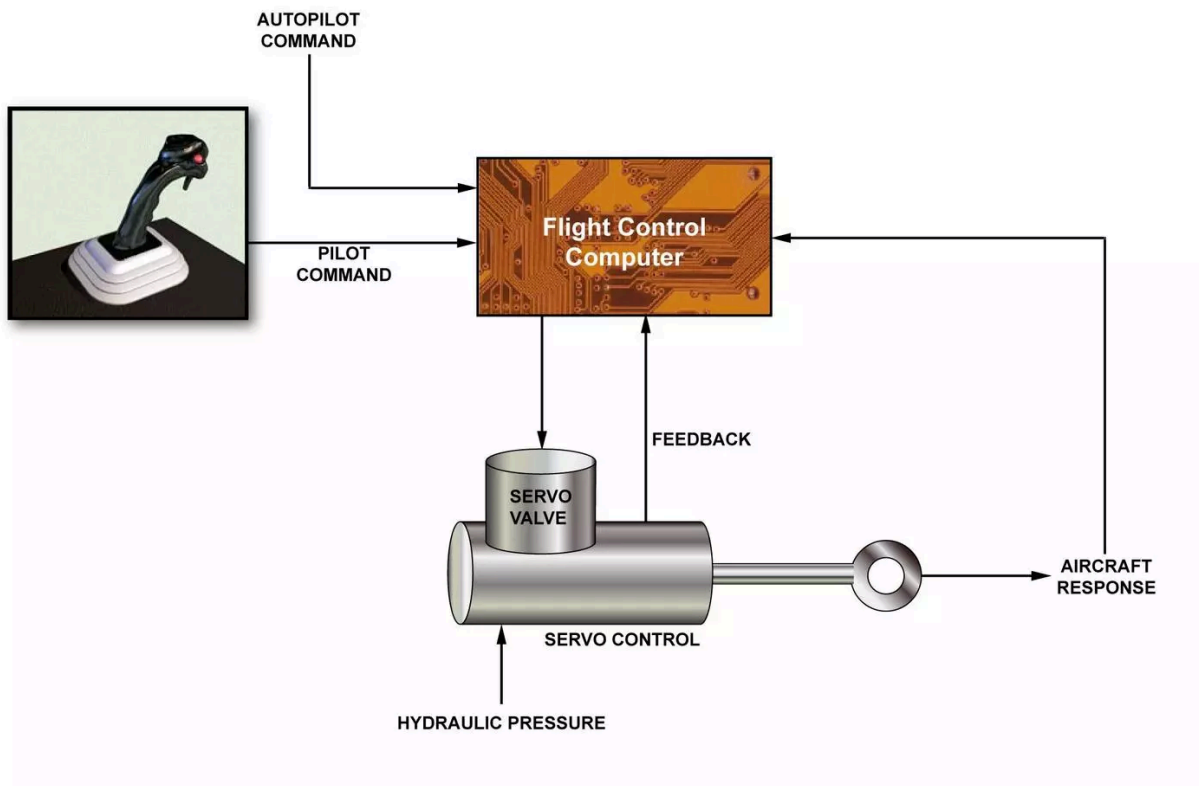
Background and Terminologies

This section separates avionics into three main sections, aviate, navigate, and communicate.

Aviate:

There are two main technologies related to Aviate, autopilot and fly-by-wire. One example of an autopilot implementation is the Boeing Honeywell Uninterruptible Autopilot, which is a system that can be used to navigate and land the aircraft autonomously and can not be deactivated by people on board. More specifically according to the patent filed by Boeing US7142971B2, the system relates to the automatic control of the path of travel of a vehicle and, in particular, to techniques for automatically controlling the path of travel of a vehicle that cannot generally be superseded or disengaged from onboard once the automatic control system is engaged. However, there are concerns over the security of the system, specifically about its design which could make it vulnerable to digital attacks.

Fly-by-wire is the flight control systems that use computers to process the flight control inputs made by the pilot or autopilot, and which send corresponding electrical signals to the flight control surface actuators. This technology replaces direct mechanical linkage. Fly-by-wire improves the safety and reliability of the aircraft by reducing human error in complicated maneuvers, simplifying the control process and ultimately leading to fewer crashes caused by human error.

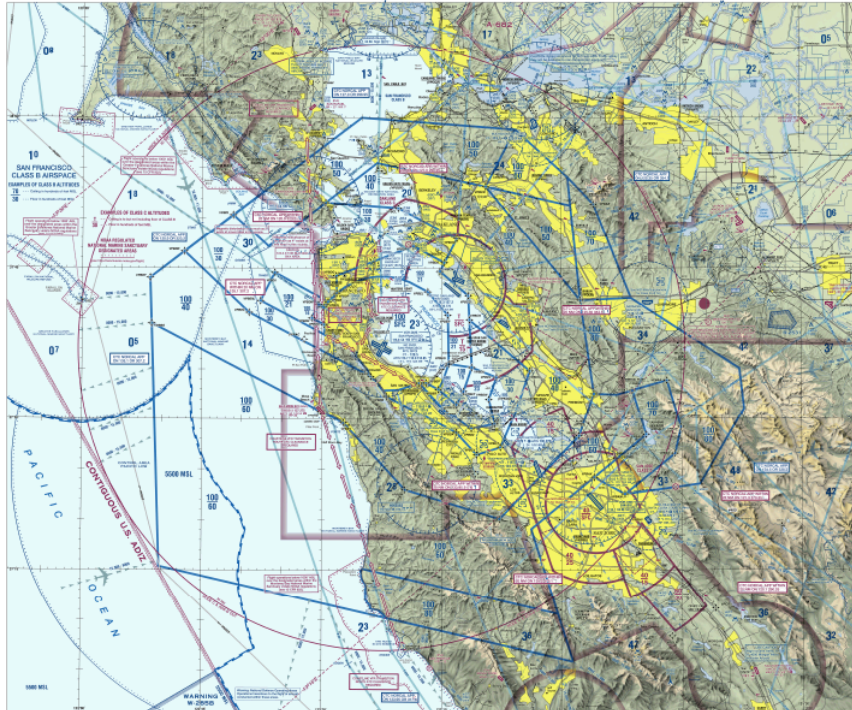


An example of a Fly-By-Wire system

Navigate:

In aviation, charts and diagrams provide critical information in supporting operations, there are three types of charts and diagrams: navigational charts, weather charts, and other miscellaneous charts.

Navigational charts like a sectional provide information to operators on information about terrain, routes, airspace, navigational aids, and other information about the airspace.



Picture to the left is an example of a Terminal Chart (TAC) of the Bay Area, which as indicated is covered by the KSFO class B airspace (In blue).

The picture below is an ILS chart for the San Francisco Airport, it provides information on terrain and navigational aids.

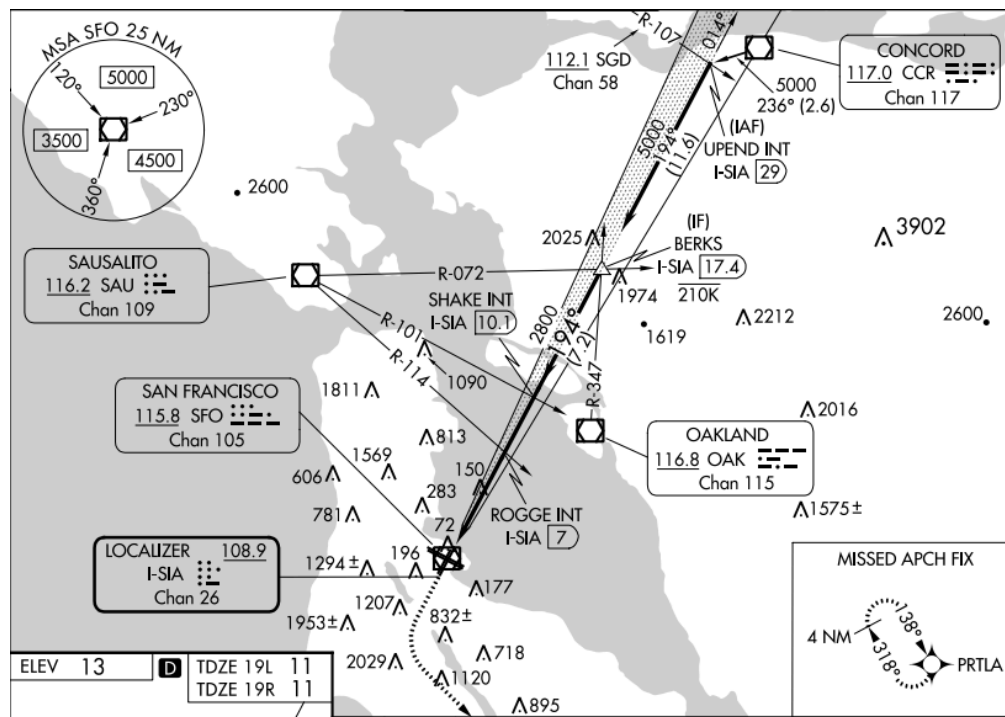


Figure 13-10 is a surface analysis chart, which is a type of weather chart.

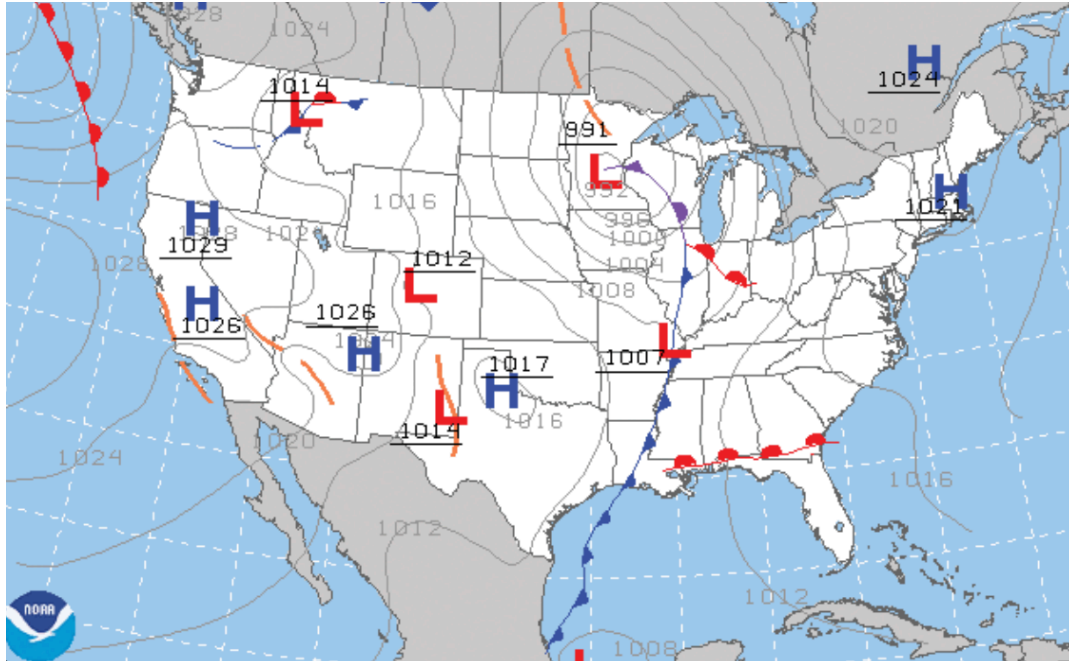


Figure 13-10. Surface analysis chart.

A weather chart is a chart that combines multiple pieces of information and provides critical meteorological information to the pilot, for example, the air pressure, wind speed, and wind direction. While weather forecasts are not 100 percent accurate, meteorologists, through careful scientific study and computer modeling, have the ability to predict weather patterns, trends, and characteristics with increasing accuracy. Through a complex system of weather services, government agencies, and independent weather observers, pilots and other aviation professionals receive the benefit of this vast knowledge base in the form of up-to-date weather reports and forecasts. These reports and forecasts enable pilots to make informed decisions regarding weather and flight safety before and during a flight.

Communicate:

Communication is an important part of safe aviation, and it has seen the most growth and development since the invention of

aviation, the most basic communication method is by radio, there are two main types of communication, with air traffic controllers or without. Without air traffic controllers pilots commonly use Common traffic advisory frequency (CTAF). Pilots use the tower frequency to coordinate their arrivals and departures safely, giving position reports and acknowledging other aircraft in the airfield traffic pattern. With air traffic controllers, all information is directed to and routed through the controller. Radio is also used to provide automated weather updates and traffic and airport information to the pilots, for example, ATIS which stands for automatic terminal information service combines the weather and NOTAMs to form a single broadcast to allow pilots to understand the situation more cohesively.



Example of a frequency chart: When there is a controller, the frequency is 118.6 MHz, when there is not a controller it is also 118.6 MHz, but pilots talk to each other instead of to the controller.

SFO ATIS INFO I 0256Z. 26020KT 10SM CLR 16/11 A3003 (THREE ZERO ZERO THREE). SIMULTANEOUS CHARTED VISUAL FLIGHT PROCEDURES IN USE. LNDG RWYS 28L, 28R. DEPG RWYS 1L, 1R. NOTAMS... TWY B1 CLSD, TWY B2 CLSD, TWY B3 CLSD. TWY B CLSD BETWEEN TWY B4, TWY Q1, TWY Q CLSD BETWEEN TWY A, TWY B.

BIRD ACTV IN VCY OF AP. ORACLE PARK TFR IN EFFECT. ...ADVS YOU HAVE INFO I.

An example of an ATIS broadcast, which is recorded on August 20, 2024, at 02:56 UTC. The first part of *ATIS SFO ATIS INFO I* indicates which airport the ATIS is providing information about, the letter *I* is a quick way for ATC to confirm that the pilot has the most up-to-date information. The second part from *26020KT 10SM to A3003 (THREE ZERO ZERO THREE)* is the weather of the airport and is provided in standard METAR format, METAR which stands for "Meteorological Aerodrome Report is a format for reporting weather information. The third part from *SIMULTANEOUS to RWYS 1L, 1R* is the current airport status and is optional and usually not included in the ATIS. The third part is from *NOTAMS... to IN EFFECT.* provides pilots with recent changes in NOTAMs, which are notices containing information essential to personnel concerned with flight operations but not known far enough in advance to be publicized by other means. It states the abnormal status of a component of the National Airspace System (NAS) – not the normal status. In this case, alerting pilots that a certain taxi way is closed¹, and more importantly, there is a temporary flight restriction (TFR), near the airport (in this case, because of a baseball game). The last part *ADVS YOU HAVE INFO I.* echos the first part and remaining pilots that when the pilot makes first contact with ATC, they should inform ATC they have information I.

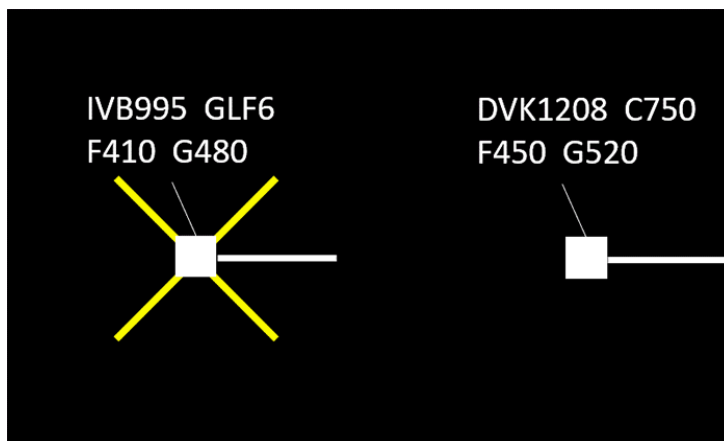
The second advancement is Radar, which has two types of

¹ Even though the NOTAMS are supposed to be signaling a abnormal status, the taxiway closures in KSFO have been broadcasted in ATIS since November 2023.

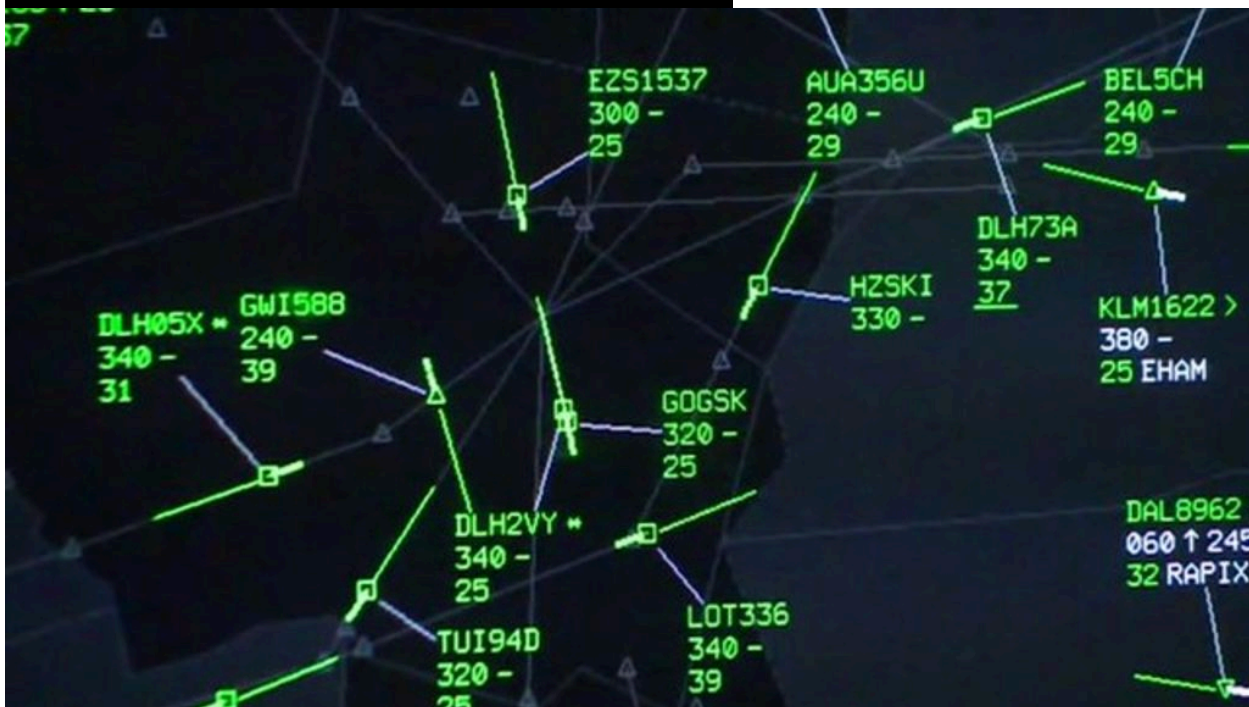
identification: primary and secondary. Radar identification is "the process of ascertaining that an observed radar target is the radar return from a particular aircraft." In other words, radar identification is confirmation that an observed target is truly the desired aircraft. In total, there are six methods for radar identification in the terminal environment (three primary methods and three secondary methods). In order to understand the different methods, we must to distinguish between the two different types of targets. A primary target is a radar target with no additional information. A secondary target is derived from information sent by a transponder. It is a radar return with other information such as altitude. A handoff is a transfer of radar contact and radio communications. In the case of an automated handoff (one received via the radar system), there has been no lapse in radar identification, so there is no need to radar identify that aircraft again.

Currently, most identifications are secondary due to the widespread use of transponders. A transponder (XPDR) is a receiver and a transmitter that will generate a reply signal upon proper request; the request and reply being on different frequencies. There are two main functions of a transponder: squawking and "IDENT". Squawking is the process of communicating between the air and the ground, the process is essential for keeping planes in the air safe and ensuring a smooth and manageable air traffic control process, both for pilots and air traffic controllers. The transponder will squawk a 4-digit code and are picked up on radar; which is used to easily communicate something about an aircraft. They can simply reveal who an

aircraft is and where they are flying, or they can relay something about a flight's situation, such as an emergency, a change in flight plan, an unexpected in-air circumstance, or other status. For example, squawking 1200 means flying VFR (visual flight rules), 7500 means hijacking, 7600 means voice radio failure, 7700 means mayday or general emergency. Another function a transponder provides is "IDENT", which just makes the aircraft highlighted or otherwise more easily noticeable on the ATC screen.



The left is what an aircraft that is flying normally looks like on the ATC's radar screen, and the right (with a yellow X) is what the aircraft looks like after pressing "IDENT"



Pictured Above: What a normal ATC's radar screen looks like.

Transponders can also provide other information such as altitude, they are classified based on their ability to transfer information.

Mode A is the most basic transponder mode. Mode A only transmits the four-digit squawk code. Useful for identifying an aircraft and its position, but not much else.

Mode C is the most widely used transponder mode. Mode C provides information on the aircraft's pressure altitude. Combined with mode A, ATC and other aircraft can receive an aircraft's unique squawk code, position, and altitude. A combination of mode A and mode C is normally just referred to as mode C.

Mode S, short for "Mode Select," is a type of transponder that offers more advanced communication capabilities than mode A or mode C transponders. A mode S transponder transmits a variety of valuable information to ATC and other aircraft. This variety of information could include:

Unique ICAO Address: Every aircraft has a unique ICAO (International Civil Aviation Organization) address assigned to it. Mode S transponders send this address, which helps ATC and other aircraft identify your specific aircraft.

Squawk Code: Just like mode A, mode S transponders also transmit the four-digit squawk code assigned by ATC.

Altitude: Similar to mode C, mode S transponders provide your aircraft's altitude.

Position, Speed, and Heading: Mode S transponders can also send your aircraft's GPS-based position, speed, and heading,

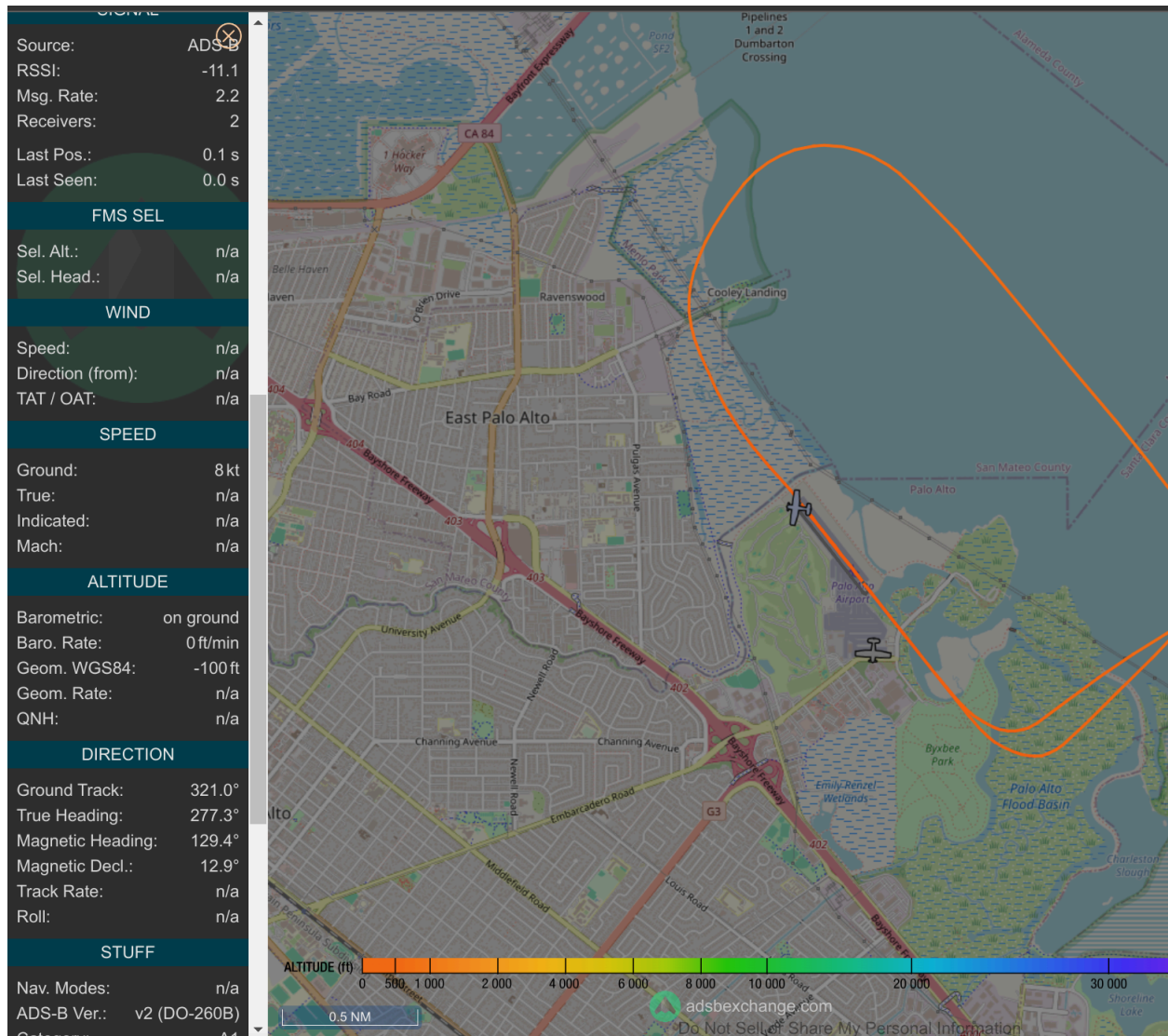
which helps ATC and nearby aircraft know your location and direction of travel.

Additional Flight-Related Data: Depending on the transponder's capabilities, it can transmit other flight-related information, such as intent data, which provides ATC with a better understanding of your planned flight maneuvers. For example, some mode S transponders can transmit the selected heading on your autopilot. If ATC notices that you have the wrong heading set, they may be able to warn the pilots.

Another advancement in the field of communication is satellites, there are two main categories of satellites the Global navigation satellite system. As of 2024, four global systems are operational: the United States's Global Positioning System (GPS), Russia's Global Navigation Satellite System (GLONASS), China's BeiDou Navigation Satellite System (BDS), and the European Union's Galileo. Those systems provide accurate location information and Time information to the pilots. The airborne radiotelephone communication via a satellite system; the use of satellites for this purpose complements satellite-based navigation capability. Aircraft onboard equipment for SATCOM includes a satellite data unit, a high-power amplifier, and an antenna with a steerable beam. A typical aircraft SATCOM installation can support data link channels for 'packet data services' as well as voice channels. SATCOM data link is currently used for only a small proportion of en route air traffic management communications in contrast to the much more extensive use as an alternative to VHF and HF for non-ATC purposes. The Asia-Pacific Region has been a particular

focus for many of the early developments in the use of SATCOM for air traffic management data link. SATCOM functionality, which primarily depends upon geostationary satellites, is poor in polar regions, where HF DL (HF Data Link) provides equivalent service for some uses. Most agencies consider that SATCOM capability will eventually form a more significant part of the ATC element of the air traffic management system, since it is probably the only way to ensure reliable data link coverage of oceanic and remote continental environments at the capacity levels required and may also eventually be required due to voice and data communication capacity constraints in parts of Europe and the USA. Current air traffic management SATCOM for en route air traffic management uses satellites and a radio frequency spectrum shared with ACARS and other communication channels rather than the dedicated ones which would be possible if the system was ultimately to be applied more widely.

The last two technologies have the same purpose, which is to provide an active transponder system that actively broadcasts the status of the aircraft instead of relying on an external signal. Automatic Dependent Surveillance-Broadcast (ADS-B) is an advanced surveillance technology that combines an aircraft's positioning source, aircraft avionics, and ground infrastructure to create an accurate surveillance interface between aircraft and ATC. ADS-B Out works by broadcasting information about an aircraft's GPS location, altitude, ground speed, and other data to ground stations and other aircraft, once per second.



Example of an aircraft equipped with ADS-B technology proving information to ATC.

Remote ID is a simplified version of an ADS-B system for drones (UAVs). It is the ability of a drone in flight to provide identification and location information that can be received by other parties through a broadcast signal.²

Past Accidents and Actions:

Korean Air Lines Flight 007

² Remote ID is only implemented in the US, and is not a global standard as other aforementioned technology.

Incident Overview:

Date: September 1, 1983

Flight: Korean Air Lines (KAL) Flight 007 (Boeing 747-230B)

Route: New York City (JFK) to Seoul (Gimpo Airport) via Anchorage

Fatalities: 269 passengers and crew, all perished

What Happened:

On September 1, 1983, KAL Flight 007, en route from Anchorage to Seoul, deviated significantly from its planned flight path and entered Soviet airspace near Sakhalin Island, a highly sensitive military zone during the height of the Cold War. The Soviet Union, perceiving the unauthorized incursion as a threat, ordered fighter jets to intercept the aircraft.

After several unsuccessful attempts to identify and communicate with the airliner, a Soviet Su-15 interceptor fired two air-to-air missiles at the Boeing 747. The missiles severely damaged the aircraft, causing it to crash into the Sea of Japan, killing all 269 people on board.

What Went Wrong:

Navigational Error, which is caused by autopilot misconfiguration. The root cause of the incident was a navigational error. The flight crew incorrectly engaged the autopilot after takeoff from Anchorage. Instead of switching the autopilot to INS (Inertial Navigation System) mode, which would have kept the aircraft on its intended course, the autopilot was left in HEADING mode. This mode maintained a constant heading instead of following the programmed course. As a result, Flight 007 gradually drifted off course, eventually straying over 200

miles away from its intended flight path into Soviet-restricted airspace.

Other contributing factors include the failure of the crew to recognize deviation, the crew of Flight 007 did not recognize that their aircraft had deviated from the intended flight path, as they failed to cross-check position indicators and alerts that might have indicated the error. The Soviet military's interception was delayed due to communication breakdowns between the ground stations and fighter pilots. This led to confusion over the nature of the aircraft, exacerbating the perception of a threat. Attempts to communicate with Flight 007 went unheeded, furthering suspicions that the aircraft was a hostile military target. It remains unclear if the flight crew received or understood these signals.

What is the solution:

The incident prompted significant improvements in international aviation safety protocols. Among the most notable was the U.S. decision to make the Global Positioning System (GPS) freely available for civilian use to prevent similar navigation errors in the future.

Current Situation & Key Issues:

On Friday, January 5, 2024, at 5:07 p.m. local time, Alaska Airlines Flight 1282 takes off from Portland International Airport for Ontario, California, with 171 passengers and six crew members on board. A few minutes after takeoff, as the plane reaches an altitude of 16,000 feet above sea level, a hole opens up next to seat 26A, prompting the pilot to declare an

emergency. Flight 1282 then initiates an emergency landing at Portland Airport. During the descent, terrified passengers captured footage showing the hole where the door plug came loose from. At 5:26 p.m. local, Flight 1282 lands, less than 20 minutes after taking off. The plane arrives at the gate at 5:34 p.m. Flight attendants instruct those who are injured to remain seated, and firefighters board the plane.

According to the National Transportation Safety Board (NTSB), the left mid-exit door (MED) plug departed the airplane during the event and the airplane returned safely to Portland International Airport. The Boeing 737-9 airplane is a twin-engine, narrow-body, transport category airplane originally certified in 2018. The airplane is equipped with a conventional tail and retractable tricycle landing gear. The airplane is 138 feet, 2 inches long, 40 feet, 10 inches tall at the tail, has a fuselage diameter of 12 feet, 4 inches, has a wingspan of 117 feet, 10 inches, and has a horizontal stabilizer span of 47 feet, 1 inch. The airplane's primary and secondary wing, fuselage, and tail structure is of all metal construction, primarily aluminum alloys. The accident airplane, SN 67501, was delivered new to Alaska Airlines on October 31, 2023, and received its airworthiness certificate on November 2, 2023. The MED plug is about 29 inches wide by 59 inches high (Figure 2). Two identical door plugs are located behind the wing, with the plug centerlines at Station 727I+2, one on each side of the fuselage. The MED plug is located and secured in the fuselage by 12 pressure stops, 2 upper guide fittings, and 2 lower hinge fittings (Figure 3). The pressure loads (radially outward) are

carried through 12 stop fittings on the MED plug, 6 on each side, that engage with 6 stop pads on a fuselage frame forward of the opening and 6 stop pads on a fuselage frame aft of the opening. The 2 upper guide fittings on the upper half of the MED plug, 1 forward and 1 aft, engage with upper guide rollers on the forward and aft fuselage frames. The 2 lower hinge fittings, 1 forward and 1 aft, installed on the lower sill engage with 2 lower hinge guide fittings on the lower end of the MED plug. The plug is secured vertically by two vertical movement arrestor bolts installed through the hinge fittings and hinge guide fittings and two upper guide track bolts installed through the upper guide fittings (Figure 4). After the vertical movement arrestor bolts and upper guide track bolts are removed, the initial MED plug motion for opening is vertically upward at a slight inclination inboard such that the plug stops will clear the fuselage stop pads. Once clear, the MED plug pivots outboard about the lower hinge fitting bolts. The MED plug is only intended to be opened for maintenance and inspection which requires removing the vertical movement arrestor bolts and upper guide track bolts. The strap assemblies below the second stop from the top will restrict the plug from opening to no further than 15° which is sufficient for routine maintenance and inspection purposes.

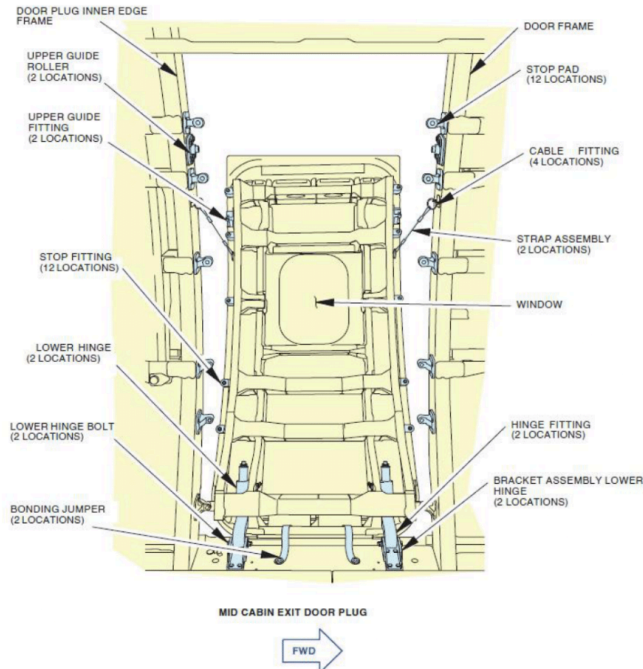


Figure 2. MED Plug drawing, open position (Source: Boeing. Image Copyright © Boeing. Reproduced with permission.)

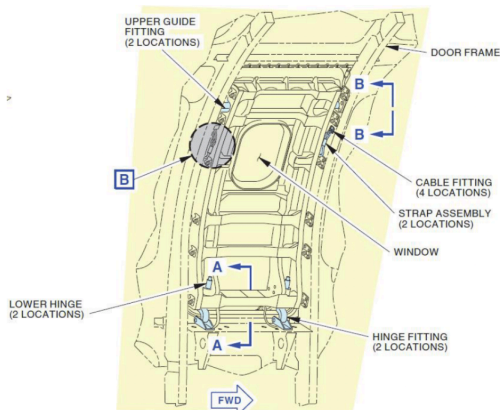


Figure 3. MED Plug drawing, closed position (Source: Boeing. Image Copyright © Boeing. Reproduced with permission.)

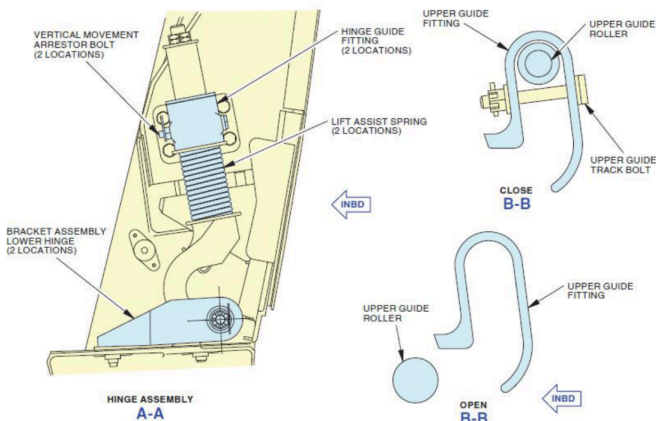


Figure 4. MED Plug lock bolt installation (Source: Boeing. Image Copyright © Boeing. Reproduced with permission.)

The FAA issued Emergency AD 2024-02-51 on January 6, 2024, based on the preliminary findings from the investigation. The AD prohibited further flight of all 737- 9 airplanes until the right and left MED plugs had been inspected and any discrepancies were corrected using a method approved by the FAA. Boeing worked with the FAA to develop the multi-operator message MOM-MOM-24-0010-01B(R4) that provided approved instructions for the inspection of the 737-9 plugs. By January 31, 2024, all but 4 airplanes had been inspected, and, in all cases, the 2 upper guide track bolts and the 2 vertical movement arrestor bolts were installed, and the plugs were engaged with the fuselage stop fittings. The 4 uninspected airplanes included the accident airplane, 2 airplanes undergoing heavy maintenance, and 1 airplane outfitted for executive transport. There were some discrepancies reported as part of the AD inspection including loose upper guide track bolts (no torque requirement existed before the MOM), the incorrect number of washers installed at various locations, clearances/gaps that were out of limits, loose fasteners at various locations, incorrect fastener grip lengths, missing cotter pins, and minor damage to components. The 2 airplanes in heavy maintenance were inspected by February 14, 2024, and found to have the upper guide track bolts and the vertical movement arrestor bolts installed. Discrepancies noted on these airplanes were similar to the others. The accident airplane's right MED plug has been inspected as outlined in section D.2.3, but the inspection of the left MED plug cannot be completed until a new left MED plug is installed. The remaining executive transport airplane is subject to United States and

European Union sanctions preventing communication with the operator.

The FAA issued Safety Alert for Operators (SAF0) 24001 on January 21, 2024, providing information on the 737-900ER MED plug. The 737-900ER MED plugs have an identical design to the 737-9 MED plugs and the FAA encouraged operators to conduct a visual inspection to ensure that the upper guide track bolts and the vertical movement arrestor bolts were installed. The available fleet data indicated that all the 737-900ER airplanes in service had surpassed the MED plug inspection threshold in the maintenance program and should have been opened and inspected at least once prior to the issuance of the SAF0.

Goals for Committee & Questions to Consider:

We could all agree that door plugs should not fall off the plane during normal flight, and nose wheels should not fall off during taxi, takeoff, and landing. Yet those situations are happening on a much more common basis.

The goal of ICAO is to impose internal and external rules and regulations to improve general operational safety and security, which could include providing funding and guidelines to improve the reliability of avionics.

Further Readings:

<https://www.nts.gov/investigations/Documents/DCA24MA063%20Preliminary%20report.pdf>

Appendix-A Positions

Argentina
Australia
Austria
Bolivia
Brazil
Canada
Chile
China
Egypt
El Salvador
Equatorial Guinea
Ethiopia
France
Germany
Ghana
Iceland
India
Italy
Jamaica
Japan
Malaysia
Mauritania
Mexico

Nigeria
Qatar
Republic of Korea
Romania
Saudi Arabia
Singapore
South Africa
Spain
United Arab Emirates
United Kingdom
United States
Venezuela
Zimbabwe

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